

# KOTIPALLI GUNA SAI

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## PROJECT EXPERIENCE

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- Arbiter (Distributed Rate Limiting & Traffic Control Platform)** – Live | GitHub | npm Jan 2026
- Designed and deployed a **distributed API traffic control platform** with **multi-cloud failover** across **2 cloud providers** to prevent abuse and ensure fair API usage.
  - Implemented **Leaky Bucket** and **Token Bucket** using **Redis atomic operations**, sustaining **1k+ req/sec** under concurrency.
  - Designed a **stateless Node.js control plane** with horizontally scalable replicas and Redis as shared coordination layer.
  - Enabled **dynamic configuration** via **web dashboard** and **code-level overrides**, supporting real-time policy updates without redeployments.
  - Published a **Node.js SDK** with **200+ npm downloads** and benchmarked **1k req/sec** at **200 ms latency**.  
(*Stack: Node.js, Redis (Upstash), Docker, NGINX, AWS EC2, GitHub Actions*)
- Conlea (Language Exchange Platform)** – Live | GitHub Aug 2025
- Built a language exchange MVP enabling **real-time chat and video calls**, used by **15+ active users**.
  - Optimized client-server interactions using **TanStack Query** for caching and background synchronization.  
(*Stack: MERN, TailwindCSS, Zustand, TanStack Query*)
- BrainWave (Cognitive Training & AI Assistant Platform)** – Live | GitHub Oct 2023 – Present
- Developed a cognitive training platform for **autism and dyslexia**, used by **10+ users**.
  - Built **4+ interactive games** targeting memory, attention, sequencing, and pattern recognition.
  - Extended the platform with a **RAG-based chatbot** using **vector retrieval** (2025), enabling domain-specific Q&A and personalized game recommendations.
  - Designed a survey-driven **early symptom detection module** achieving **90% accuracy** on curated datasets.
  - Led a team of **6 members**, managing architecture, feature development, and full-stack integration.  
(*Stack: React, Node.js, MongoDB, Flask, LLMs, Vector DB*)

## TECHNICAL SKILLS

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**Languages:** Java, python, TypeScript

**Frameworks & Libraries:** React, Node.js, Express.js, Flask

**Web & API Technologies:** HTML, CSS, REST APIs, TailwindCSS, Socket.io

**Databases & Storage:** MongoDB, MySQL, Object Storage (AWS S3, MinIO)

**Cloud & DevOps:** AWS (EC2, S3, VPC), Docker, Kubernetes

**Core CS Concepts:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

**Tools:** Git, GitHub, VS Code, PyCharm, Postman

## EDUCATION

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|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Keshav Memorial Institute of Technology, Hyderabad, India</b><br><i>Bachelor of Technology in Computer Science and Engineering</i> | Nov 2022 – July 2026<br><b>CGPA: 9.03 / 10.0</b> |
| <b>Bhashyam Junior College, Hyderabad, India</b><br><i>Intermediate (11<sup>th</sup> &amp; 12<sup>th</sup> Grade) – MPC</i>           | Jun 2020 – Apr 2022<br><b>Score: 98.1%</b>       |
| <b>Bhashyam Public School, Hyderabad, India</b><br><i>Secondary Schooling (10<sup>th</sup> Grade) – SSC</i>                           | Jun 2019 – Apr 2020<br><b>GPA: 10.0 / 10.0</b>   |

## ACHIEVEMENTS

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- Winner, **Centific Premier Hackathon 2.0 (AI-Systems-Engineering stream)**.
- Achieved a **LeetCode Contest Rating of 2078 (Knight)**, ranking in the **top 1.72%** globally.